

EE3621: Digital Signal Processing

Lecture 3: The Discrete-Time Fourier Transform (DTFT)

(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Motivation: Continuous spectrum for discrete signals and periodic spectrum.
 - **05:00 – 12:00 (7 mins):** DTFT & IDTFT Definitions and Convergence Criteria.
 - **12:00 – 22:00 (10 mins):** DTFT of Common Signals: Rectangular Pulse, Decaying Exponential, and Double-Sided Exponential.
 - **22:00 – 35:00 (13 mins):** Key Properties of the DTFT with full algebraic proofs.
 - **35:00 – 40:00 (5 mins):** Duality, the Uncertainty Principle, and Checkpoint.
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2 Mathematical Definition of DTFT

The Discrete-Time Fourier Transform (DTFT) maps a sequence $x[n]$ to a periodic frequency function:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1)$$

The inverse transform (IDTFT) is:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \quad (2)$$

The DTFT converges uniformly if $x[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty \quad (3)$$

3 DTFT of Common Signals

3.1 Finite Rectangular Pulse (Length M)

For $x[n] = 1$ for $0 \leq n \leq M-1$:

$$X(e^{j\omega}) = \sum_{n=0}^{M-1} e^{-j\omega n} = \frac{1 - e^{-j\omega M}}{1 - e^{-j\omega}} = e^{-j\omega(M-1)/2} \frac{\sin(\omega M/2)}{\sin(\omega/2)} \quad (4)$$

3.2 Single-Sided Decaying Exponential

For $x[n] = a^n u[n]$ ($|a| < 1$):

$$X(e^{j\omega}) = \sum_{n=0}^{\infty} (ae^{-j\omega})^n = \frac{1}{1 - ae^{-j\omega}} \quad (5)$$

3.3 Double-Sided Decaying Exponential

For $x[n] = a^{|n|}$ ($|a| < 1$):

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \frac{1 - a^2}{1 + a^2 - 2a \cos \omega} \quad (6)$$

4 Key Properties of the DTFT

For $x[n] \leftrightarrow X(e^{j\omega})$ and $y[n] \leftrightarrow Y(e^{j\omega})$:

4.1 Time Shifting

$$x[n - n_0] \longleftrightarrow X(e^{j\omega}) e^{-j\omega n_0} \quad (7)$$

Proof: Let $m = n - n_0$.

$$\text{DTFT}\{x[n - n_0]\} = \sum_{n=-\infty}^{\infty} x[n - n_0] e^{-j\omega n} = \sum_{m=-\infty}^{\infty} x[m] e^{-j\omega(m+n_0)} = X(e^{j\omega}) e^{-j\omega n_0} \quad (8)$$

See Figure 1 for magnitude and phase profiles under time delay.

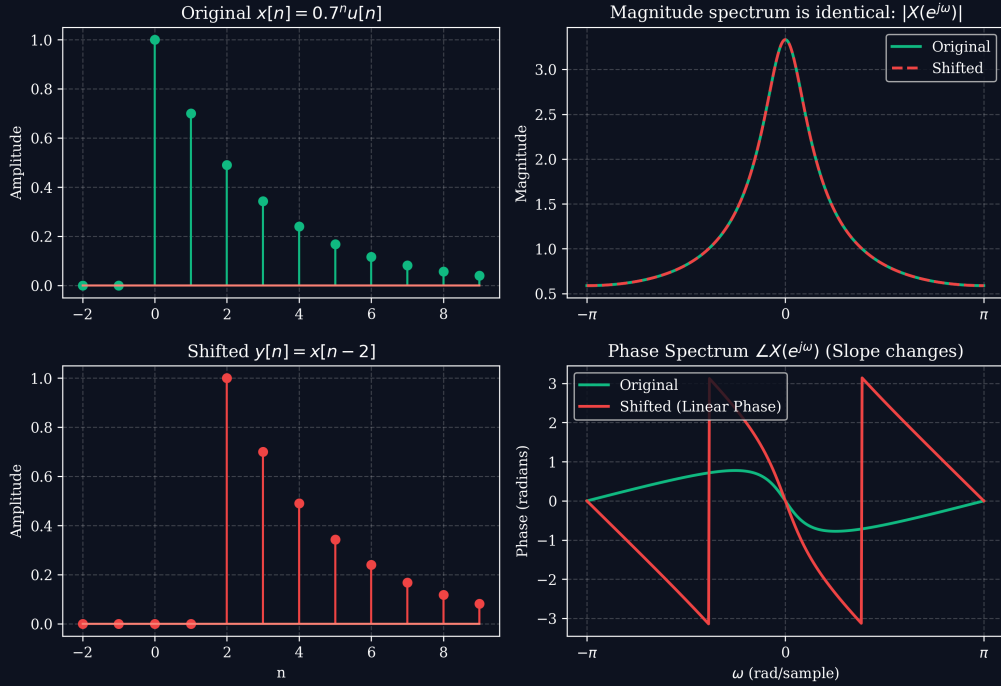


Figure 1: Time Shifting property: magnitude response remains invariant while phase shifts linearly.

4.2 Frequency Shifting (Modulation)

$$x[n]e^{j\omega_0 n} \longleftrightarrow X(e^{j(\omega-\omega_0)}) \quad (9)$$

4.3 Differentiation in Frequency

$$nx[n] \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega} \quad (10)$$

Proof:

$$\frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} x[n](-jn)e^{-j\omega n} \implies j \frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} (nx[n])e^{-j\omega n} \quad (11)$$

4.4 Convolution Property

$$x[n] * h[n] \longleftrightarrow X(e^{j\omega})H(e^{j\omega}) \quad (12)$$

Proof:

$$\sum_n (x[n] * h[n])e^{-j\omega n} = \sum_n \sum_k x[k]h[n-k]e^{-j\omega n} = \sum_k x[k]e^{-j\omega k} \sum_m h[m]e^{-j\omega m} = X(e^{j\omega})H(e^{j\omega}) \quad (13)$$

4.5 Parseval's Relation

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega \quad (14)$$

5 Duality & The Uncertainty Principle

A signal cannot be narrow in both time and frequency domains simultaneously (Figure 2).

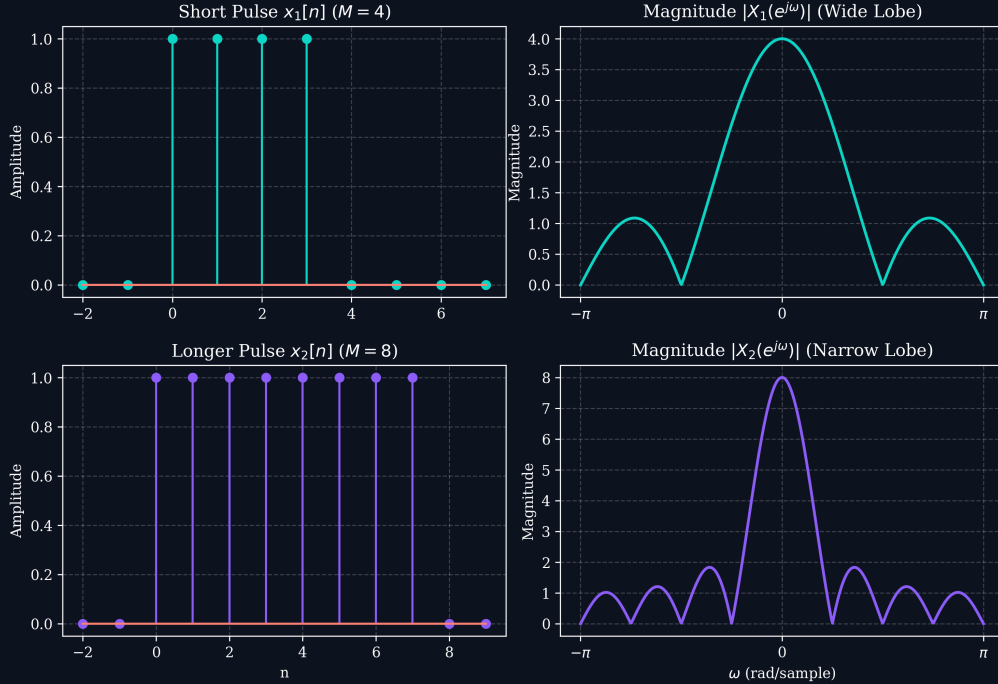


Figure 2: Duality: shorter rectangular time pulse maps to a wider frequency response main lobe.